

QCE Chemistry Units 3&4

Test 1: Chemical equilibrium systems

- Chemical equilibrium
- Factors that affect equilibrium
- Equilibrium constants
- Properties of acids and bases
- pH scale

SECTION 1 – MULTIPLE CHOICE QUESTIONS

QUESTION 1 D

D is correct. For the equilibrium equation for a reaction $aA + bB \rightleftharpoons cC + dD$, the formula for the

equilibrium expression is $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$. The product concentrations are on the top (numerator) and

the reactant concentrations are on the bottom (denominator). Each product and reactant must be raised to the power of its coefficient as shown in the equilibrium equation.

A is incorrect. This option shows the reactant concentration as the numerator.

B is incorrect. This option shows the reactant concentration as the numerator, and the reactant and products are not raised to the power of their coefficients.

C is incorrect. The reactant and product concentrations are not raised to the power of their coefficients.

QUESTION 2 B

B is correct, and **A** and **D** are incorrect. A small equilibrium constant (K_c) indicates that there is a larger concentration of reactants than products at equilibrium.

C is incorrect. If a reaction is at equilibrium, the rates of the forward and reverse reactions are the same.

QUESTION 3 C

C is correct. Changes in pressure only affect systems where, at equilibrium, the number of gaseous reactant particles and the number of gaseous product particles are different.

A and **B** are incorrect. From the equilibrium equation, it can be determined that the number of gaseous reactant particles and gaseous product particles are the same. Therefore, the system will not be affected by a decrease in pressure.

D is incorrect. The system will still be at equilibrium.

QUESTION 4 C

$$\begin{aligned}n(\text{H}_2 \text{ reacted}) &= 15 - 12 \\&= 3 \text{ mol}\end{aligned}$$

$\text{H}_2 : \text{N}_2 = 3 : 1$, therefore:

$$\begin{aligned}n(\text{N}_2 \text{ reacted}) &= n(\text{H}_2) \times \frac{1}{3} \\&= 3 \times \frac{1}{3} \\&= 1 \text{ mol}\end{aligned}$$

$$\begin{aligned}n(\text{N}_2 \text{ unreacted}) &= 15 - 1 \\&= 14 \text{ mol}\end{aligned}$$

$\text{H}_2 : \text{NH}_3 = 3 : 2$, therefore:

$$\begin{aligned}n(\text{NH}_3 \text{ produced}) &= n(\text{H}_2) \times \frac{2}{3} \\&= 3 \times \frac{2}{3} \\&= 2 \text{ mol}\end{aligned}$$

QUESTION 5 D

D is correct. As the K_c of a reaction is determined at specific temperatures, the value of the K_c will be affected by a change in temperature.

A is incorrect. Adding a catalyst does not affect a system at equilibrium as the ratio between reactants and products does not change.

B and **C** are incorrect. Changes in pressure or volume do not change the kinetic energy of the particles and, therefore, do not change the K_c value.

QUESTION 6 D

For each decrease of 1 point on the pH scale, there is an increase in the hydrogen ion (H^+) concentration by a factor of 10. A pH difference of 2 (3 to 5) is, therefore, an decrease of 100 times (10×10) the H^+ concentration.

QUESTION 7 B

The H^+ concentration can be calculated as $10^{-\text{pH}}$; therefore, $10^{-2.8} = 1.6 \times 10^{-3}$.

QUESTION 8 C

$$\text{pOH} + \text{pH} = 14$$

$$\text{pOH} = 14 - \text{pH}$$

$$= 14 - 6$$

$$= 8$$

QUESTION 9 B

B is correct. A reaction at equilibrium does not need to contain equal amounts of reactants and products.

A is incorrect. A reaction at equilibrium does not change the concentrations of reactants or products, so a constant colour is likely to be observed.

C is incorrect. Chemical equilibrium is a dynamic equilibrium, so reactions are still occurring at equilibrium.

D is incorrect. Equilibrium is temperature dependent, so a constant temperature is maintained at equilibrium.

QUESTION 10 A

The first reaction has been doubled and reversed. This means that the K_c (1000) of the first reaction needs to be squared and then inversed to find the K_c of the second reaction. Therefore, $K_c = \frac{1}{1000^2} = 1 \times 10^{-6}$.

SECTION 2

QUESTION 11 (5 marks)

- a) The forward reaction is exothermic.

Any one of:

- As the temperature increases, the percentage of product decreases. The system opposes an increase in temperature by favouring the reverse, endothermic reaction.
- As the temperature decreases, the percentage of product increases. The system opposes a decrease in temperature by favouring the forward, exothermic reaction.

*[3 marks]**1 mark for stating that the forward reaction is exothermic.**1 mark for explaining how the temperature affects the percentage of product.**1 mark for explaining how the system responds to a change in temperature.*

- b) The reactants have more gaseous particles than the products.

Any one of:

- As the pressure increases, the percentage of product increases. An increase in pressure causes the system to respond to reduce the pressure by favouring the forward reaction, which produces fewer gaseous particles in the products.
- As the pressure decreases, the percentage of product decreases. A decrease in pressure causes the system to respond to increase the pressure by favouring the reverse reaction, which produces more gaseous particles in the reactants.

*[2 marks]**1 mark for identifying that the reactants have more gaseous particles.**1 mark for explaining how the system responds to a change in pressure.*

QUESTION 12 (4 marks)

$$K_c = \frac{[\text{Cl}_2][\text{NO}]^2}{[\text{NOCl}]^2}$$

$$= \frac{3.22 \times 2.65^2}{0.067^2}$$

$$= 5037.3 \dots$$

As the two reactions occur at the same temperature, K_c is the same. Thus, finding the concentration of NOCl in the second vessel gives:

$$K_c = \frac{[\text{Cl}_2][\text{NO}]^2}{[\text{NOCl}]^2}$$

$$5037.30 \dots = \frac{2.76 \times 1.79^2}{[\text{NOCl}]^2}$$

$$[\text{NOCl}] = \sqrt{0.0017 \dots}$$

$$= 0.0419 \text{ mol L}^{-1}$$

*[4 marks]**1 mark for substituting the concentrations in the first vessel into the K_c formula.**1 mark for calculating the equilibrium constant.**1 mark for substituting the concentrations in the second vessel into the K_c formula.**1 mark for determining the concentration of NOCl in the second vessel.*

QUESTION 13 (5 marks)

a) $K_w = [\text{H}^+][\text{OH}^-] = [\text{H}^+]^2$

$$\text{pH} = -\log \sqrt{K_w}$$

$$\begin{aligned}\text{pH at } 30^\circ\text{C} &= -\log \sqrt{1.47 \times 10^{-14}} \\ &= 6.92\end{aligned}$$

$$\begin{aligned}\text{pH at } 100^\circ\text{C} &= -\log \sqrt{51.3 \times 10^{-14}} \\ &= 6.14\end{aligned}$$

[3 marks]

*1 mark for substituting K_w into the formula for pH.**1 mark for calculating the pH at 30°C.**1 mark for calculating the pH at 100°C.*

- b) The pH of water is lower at 100°C (6.14) than at 30°C (6.92), meaning that the concentration of H^+ in water is higher at 100°C. At 100°C, the system opposes the increase in temperature by favouring the forward reaction (the ionisation of water), which produces H^+ . Therefore, the ionisation of water is endothermic.

[2 marks]

*1 mark for explaining that the system favours the forward reaction at higher temperatures with reference to the pH of water.**1 mark for stating that the ionisation of water is endothermic.**Note: Consequential on answer to **Question 13a**). Responses must refer to the answer obtained in **13a**).***QUESTION 14 (1 mark)**

A weak acid does not completely dissociate, so the concentration of the acid is not equal to the concentration of H^+ .

[1 mark]

*1 mark for explaining that concentration of a weak acid is not equal to the concentration of H^+ due to incomplete dissociation.***QUESTION 15 (2 marks)**

In an open container, the carbon dioxide gas produced can continue to escape, meaning that the reverse reaction does not occur and complete decomposition does occur.

In a sealed container, the carbon dioxide gas produced remains in the container, allowing the reverse reaction to occur and equilibrium to be established.

[2 marks]

*1 mark for explaining why complete decomposition occurs in an open container.**1 mark for explaining why equilibrium is established in a sealed container.*

QUESTION 16 (4 marks)

Experiment 1: $\frac{2 \times 2}{1 \times 1} = 4$

Therefore, the reaction in experiment 1 did not reach equilibrium. (*The number obtained is less than 6, meaning more products are needed.*) The reaction would need to proceed to the right (the forward reaction) to reach equilibrium.

Experiment 2: $\frac{3 \times 4}{4 \times 0.5} = 6$

Therefore, the reaction in experiment 2 reached equilibrium.

Experiment 3: $\frac{5 \times 3}{1.5 \times 1} = 10$

Therefore, the reaction in experiment 3 did not reach equilibrium. (*The number obtained is greater than 6, meaning more reactants are needed.*) The reaction would need to proceed to the left (the reverse reaction) to reach equilibrium.

Experiment 4: $\frac{3 \times 2}{0.5 \times 2} = 6$

Therefore, the reaction in experiment 4 reached equilibrium.

[4 marks]

1 mark for determining that the reaction in experiment 1 did not reach equilibrium and would need to proceed to the right.

1 mark for determining that the reaction in experiment 2 did reach equilibrium.

1 mark for determining that the reaction in experiment 3 did not reach equilibrium and would need to proceed to the right.

1 mark for determining that the reaction in experiment 4 did reach equilibrium.

Note: The state (aq) has been used to ensure that responses include all values in calculations. Responses must show working to obtain full marks but do not need to explain the significance of the numbers obtained.

QUESTION 17 (3 marks)

$$K_c = \frac{[\text{Br}_2][\text{Cl}_2]}{[\text{BrCl}]^2}$$

$$[\text{Br}_2] = [\text{Cl}_2] \\ = 4.60 \text{ mol}$$

$$[\text{BrCl}]^2 = \frac{4.60 \times 4.60}{13.7}$$

$$[\text{BrCl}] = \sqrt{\frac{4.60 \times 4.60}{13.7}} \\ = 1.24$$

[3 marks]

1 mark for determining the amount of Br_2 .

1 mark for determining the amount of BrCl .

1 mark for showing working.

QUESTION 18 (4 marks)

- a) The equilibrium position would shift to the right/the forward reaction would be favoured.
(According to Le Chatelier's principle, an increase in reactants would result in the system wanting to use up these additional reactants, and overall the system would shift forward or to the right.)
[1 mark]
1 mark for stating that the equilibrium position would shift to the right.
- b) The equilibrium position would not change. (While a catalyst may speed up the reaction rate, it would not alter the position of equilibrium and hence there would be no changes.)
[1 mark]
1 mark for stating that the equilibrium position would not change.
- c) The equilibrium position would shift to the left/the reverse reaction would be favoured. (A decrease in volume results in an increased pressure of the system. According to Le Chatelier's principle, the system would want to move to a position to decrease the pressure. In this system, there are fewer particles on the left; hence the system will favour the reverse reaction or move to the left.)
[1 mark]
1 mark for stating that the equilibrium position would shift to the left.
- d) The equilibrium position would shift to the right/the forward reaction would be favoured.
(According to Le Chatelier's principle, the system would need to move to remove the additional energy that has been introduced by a temperature increase. Since this is an endothermic reaction, the system would move to the product side to absorb the additional energy. Therefore, the equilibrium position would shift to the right.)
[1 mark]
1 mark for stating that the equilibrium position would shift to the right.

QUESTION 19 (2 marks)

Both pure liquids have no (or very few) charged particles (ions), which allow electrical conductivity.

When sulfuric acid ($\text{H}_2\text{SO}_4(\text{aq})$) is dissolved in water ($\text{H}_2\text{O}(\text{l})$), it dissociates into H^+ and sulfate ions (SO_4^{2-}), which allows electricity to be conducted.

[2 marks]
*1 mark for explaining why the pure liquids are poor conductors.
1 mark for explaining why a mixture of H_2SO_4 and H_2O is a good conductor.*